Getting Started with OpenDaylight: Theory

1. How to Get Started

- **Download & Installation:** OpenDaylight (ODL) can be downloaded from its official website as a distribution or built from source.
- **Running the Controller:** You start ODL by launching its Karaf container (a modular OSGi runtime).
- **Using CLI and GUI:** ODL offers a CLI for management and a web-based GUI (DLUX) to interact with network applications.
- **Adding Features:** ODL uses feature management to add components (e.g., OpenFlow plugin).

2. OpenDaylight Clustering

- Clustering allows multiple ODL controller instances to run together for **scalability** and **high availability**.
- Controllers share the network state using distributed data stores, ensuring consistency.

- Clustering supports fault tolerance by allowing failover between nodes.

3. Model-Driven Service Abstraction Layer (MD-SAL)

- MD-SAL is the **core data management and service abstraction** layer in ODL.
- It uses **YANG models** to define network data and services.
- MD-SAL provides a **consistent and unified API** for applications to interact with network state.
- Supports **data stores**: configuration, operational, and candidate.

4. Internal Datastore

- ODL maintains **separate datastores**:

- **Config datastore:** Stores intended network configurations.
- **Operational datastore:** Reflects the current state of the network.

- These datastores are synchronized with network devices via southbound protocols.

5. OpenFlow Plugin

- This plugin enables ODL to communicate with OpenFlow-enabled switches.
- It translates flow rules and other instructions into OpenFlow messages.
- Manages flow tables, port status, and statistics.

6. Open vSwitch (OVS) Concepts

- OVS is a **software-based OpenFlow switch** widely used for SDN experiments.
- Runs on Linux and supports OpenFlow protocol.
- It allows creation of virtual switches in virtualized environments like Mininet.

7. Mininet Overview

- Mininet is a **network emulator** that creates a virtual network with hosts, switches, and links on a single machine.
- Used for SDN development/testing by running real code (like ODL) with virtual switches (e.g., OVS).
- Easy to start and reset virtual topologies.

8. L2Switch Application

- A simple ODL network application that provides **basic Layer 2 switching**.
- It learns MAC addresses and programs flow entries to switch packets.
- Useful for learning and testing SDN fundamentals.